National University of Lesotho
Department of Physics and Electronics

EE3304 - Electrical Circuit Analysis

Lab 03: Power Analysis in AC Circuits

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1 Lab Brief and Purpose

Power is the most important quantity in electronic systems, because such systems involve transmission of power from one point to another. Also, every household electrical device - e.g. a fan, TV, bread maker, coffee maker, iron, etc. - has a power rating that indicates how much power the device requires; exceeding the power rating can permanently destroy an appliance. In this lab, you will perform power analysis and design in AC circuits. You should complete the pre-lab exercises before coming to the lab.

2 Pre-Lab Exercises

1. The network shown in Figure 1 models the power transmission system.
   (a) Find the total complex power S_{T} supplied by the source $\mathrm{V_s}$, the complex power absorbed by the "transmission line" S_{line} and the complex power absorbed by the "load" S_{L}. Assume a sinusoidal source with 10 $\mathrm{V_{pp}}$ and 1 kHz frequency. Use rms values for all the calculations.
   (b) Calculate the load power factor (pf).

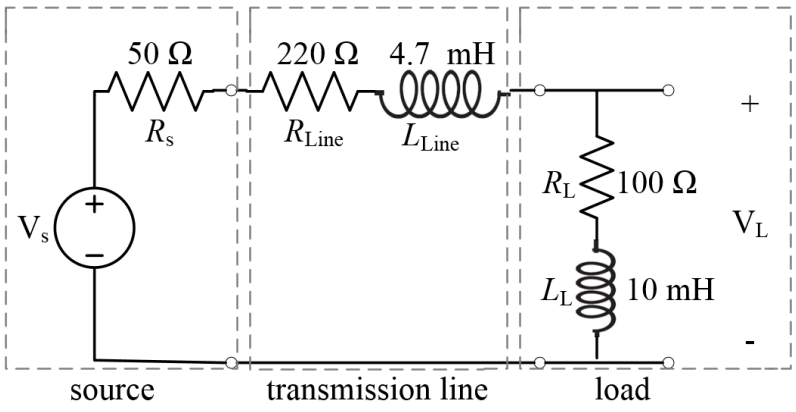


Figure 1: Power Distribution Network

2. Compute the capacitance that must be placed in parallel with the load to correct the power factor to unity.
3. Re-design a load network that will maximize the average power drawn from the network. Your design should extract at least 95%. (i.e. *you are basically determining the values of R_L, L_L and C_L components you will use in the lab*).

3 In-Lab Exercises

1. Set up an experiment as shown in the circuit diagram of Figure 1.
   (a) Set the signal generator to produce a sinusoidal with a frequency of 1 kHz and an amplitude of 10 $\mathrm{V_{pp}}$.

(b) Measure the amplitude of both v_s and v_L, and also measure the phase difference between v_s and v_L.

(c) Measure the amplitude and phase of the current i_L. i.e. measure the amplitude and phase of the voltage across the load resistor, 10 kΩ. This helps in calculating the amplitude and phase angle of the current i_L.

2. Construct the network with the capacitor predicted in Pre-Lab Exercise 2 included.

   (a) Measure the amplitude and phase of both v_L and i_L and verify that the power factor has been corrected.

3. Build the power transmission system with the load network developed in Pre-Lab Exercise 3.

   (a) Perform appropriate measurements to verify that your load circuit maximizes power transfer as you predicted. The efficiency of the power system can be evaluated by comparing the amount of average power delivered by the source, with the amount of average power dissipated in the load.

4 Post-Lab Exercises

1. Refer to your measurements from In-Lab Exercise 1. Use your measurements to determine the complex power S_T and S_L. Compare these results with the complex power calculated in your pre-lab exercise. Also, from these measurements, compute the load power factor. Explain any discrepancies.

2. Refer to your measurements from In-Lab Exercise 2. Verify that the power factor has been corrected. Comment on any discrepancies.

3. Refer to your measurements from In-Lab Exercise 3. Use your measurements to compute the efficiency of your system. Explain any discrepancies.

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